Central & Spherical Projection: Geometry, the Multi-Axis Model, and a Glance at a Physical Stage (Observational Note)

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Observational note. No new geometric theorem is claimed. A classical map (radial projection) is organized, and the structures appearing on it are observed. No physical derivation is made.

1. Spherical projection σ_R (radial projection)

$$\sigma_R : \mathbb{R}^{n+1} \setminus \{0\} \to S^n(R), \qquad \sigma_R(x) = \frac{R}{\|x\|}\, x$$

- Identical to the **radial projection** (the deformation retract of Hatcher, *Algebraic Topology*, Ch.0). A C^∞ retraction, idempotent $\sigma_R \circ \sigma_R = \sigma_R$.
- Quotient $(\mathbb{R}^{n+1} \setminus \{0\})/\mathbb{R}_{>0} \cong S^n(R)$. $\ker D\sigma_R =$ radial span$\{x\}$; $\operatorname{im} D\sigma_R = x^\perp = T_{\sigma_R(x)}S^n(R)$ (a submersion of rank n). Angle-preserving; direction invariant under the radius scale.

2. Central (gnomonic) projection $\Phi_R = \sigma_R|_{\Pi_R}$

Restriction to the tangent hyperplane $\Pi_R = \{x_{n+1} = R\} \cong \mathbb{R}^n$:

$$\Phi_R = \sigma_R\big|_{\Pi_R} : \ \Pi_R \ \xrightarrow{\ \sim\ }\ S^n_+(R) \quad \text{(open upper hemisphere; diffeomorphism)}$$

- The pullback metric $g_{\mu\nu} = \dfrac{R^2}{\ell^2}\Big(\delta_{\mu\nu} - \dfrac{x_\mu x_\nu}{\ell^2}\Big)$ ($\ell = \sqrt{R^2 + |x|^2}$) satisfies $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ ($\Lambda = \frac{(n-1)(n-2)}{2R^2}$), coinciding intrinsically with the **Beltrami coordinates of de Sitter space**. As $R \to \infty$ it degenerates to the flat (Minkowski) case.
- **The crux**: the contrast between σ_R (non-injective, whole sphere) and Φ_R (injective, upper hemisphere only).

3. The multi-axis model

Choosing any of the $n+1$ axes of the background $\mathbb{R}^{n+1}$ as the projection center yields the same metric/curvature structure (**axis equivalence**); $n+1$ subjective spaces coexist, with transitions $T_{A\to B} = \Phi_B^{-1} \circ \Phi_A$. The roles of the "projection-center axis" (inaccessible from inside) and a "subjective coordinate axis" interchange with the observer.

4. A glance at a physical stage (observation only)

Taking the central projection of a vacuum (an exterior-free intrinsic system) as the homogeneous 4-sphere $S^4(R_\mathrm{U})$ (sectional curvature $1/R_\mathrm{U}^2$, the Euclidean version of dS_4), one observes that "phase-bearing local modes (particle-like states)" $P_a = (\boldsymbol{\nu}_a, R_a)$ with spread $W_a = 2R_a$ can be placed on it. No Lorentzian metric, causal structure, or physical derivation is claimed. [Paper 14]

References (Concept DOI)

- Definition of the spherical projection σ_R (foundational map): 10.5281/zenodo.20462569
- A geometric formulation of 4-dimensional space by central projection: 10.5281/zenodo.19427780
- Composition of central projections (multi-axis; closed form of the composite curvature radius): 10.5281/zenodo.20060728
- [Paper 14] Central projection of a vacuum universe and particle-like states with a spread phase: 10.5281/zenodo.20543044

GitHub: github.com/WurabeSeiji/ai-chat-logs-open / Full texts on Zenodo (JA/EN, CC BY 4.0)